Manthan Chawda

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Professional Summary

Computer Science undergraduate with interests in AI/ML, mathematics and software development. Experienced in building practical projects across machine learning, computer vision, software systems, robotics and drones. Interested in applying strong technical foundations to real-world problems while exploring research and new areas of computing.

Education

GSFC University — Vadodara, Gujarat
B.Tech in Computer Science & Engineering (AI/ML Specialization) — CGPA: 8.47 — *July 2024 – Present*

Bright School — Vadodara, Gujarat
Higher Secondary & Secondary Education, PCM — *May 2024*

Achievements

First Runner-Up – ELCIA Next-Gen Innovative Tech Hackathon 2026
Smart City Drone-AI Challenge – Team CivicPulse, among 183 teams from 18 states and 100+ institutions.
Top 20 – TetraTHON 2026 – *Indo-French AI Innovation Sprint*, Team InvoiceGuard AI.

Certifications

TIDES Ignition Program 2026 – IIT Roorkee — *January 2026*
Oracle Cloud Infrastructure Data Science Professional – Oracle — *October 2025*
Oracle Cloud Infrastructure AI Foundations – Oracle — *October 2025*
AWS Cloud Foundations – Amazon Web Services — *July 2025*

Experience

Academic Associate — January 2026 – Present
GSFC University — *Vadodara, Gujarat*

- Guide undergraduate students through experimental physics laboratory sessions, explaining concepts and helping bridge the gap between theory and practical application.
- Set up and maintain laboratory equipment and supervise hands-on experiments while ensuring proper equipment handling and adherence to laboratory safety procedures.

Intern — December 2025
GUIITAR Council, GSFC University — *Vadodara, Gujarat*

- Researched aerodynamic and modular drone designs inspired by mini-FPV platforms, with a focus on improving flight efficiency and range.
- Analyzed and proposed structural design iterations for lightweight 3D-printed wings and body components to improve aerodynamics and flight duration.

Secretary, Entrepreneurship Club — August 2025 – September 2026
GSFC University — *Vadodara, Gujarat*

- Coordinate student-led entrepreneurship events, guest sessions and club activities, including planning, documentation and logistics.
- Prepare meeting agendas, maintain club records and coordinate communication among students, mentors and club members while supporting event execution.

Treasurer, Heritage Club — August 2025 – August 2026
GSFC University — *Vadodara, Gujarat*

- Manage club finances, maintain financial records and oversee budgets and expenditures for cultural and heritage activities.
- Collaborate with the executive committee to plan budgets and track expenditures for club initiatives and events.

Projects

CivicPulse | *YOLO11m-seg, ByteTrack, MiDaS, Python, PyTorch, FastAPI* August 2026 – Present

- Developed an **AI-powered civic infrastructure monitoring platform** that analyses drone footage to detect potholes, waterlogging, open manholes, drainage overflow, and damaged footpaths.
- Built a computer vision pipeline using **YOLO11m-seg for detection and segmentation**, ByteTrack for tracking hazards across video frames, and MiDaS for depth estimation.
- Designed an **incident-based workflow** that combines repeated detections into persistent civic incidents and assigns severity and priority using defined scoring logic.
- Integrated **FastAPI, React/Vite and Supabase/PostgreSQL** to process detections, store incident data, associate telemetry, and present results through an interactive dashboard.

Code2Edge – IBM Bob 2.0 | *Edge AI, KWS, C/C++, STM32U585* September 2026

- Developing a **repository-to-edge deployment workflow** that traces ML pipelines, generates embedded implementations, and validates **numerical parity** between host/reference and MCU inference.
- Targeting the **STM32U585 MCU** on Arduino UNO Q with **quantized keyword-spotting inference**, TensorFlow Lite Micro (TFLite Micro), and optimized Arm MCU kernels.
- Structuring the deployment gate as **generated C/C++ → host parity test → MCU build/flash → on-device comparison and benchmarking**.

InvoiceGuard AI | *Python, FastAPI, React, Gemini Vision, SQLite, Docker* August 2026

- Developed an **AI-powered invoice auditing pipeline** using Gemini Vision for data extraction, with automated **vendor validation, duplicate detection, and financial risk assessment**.
- Implemented **ledger matching, GST validation, and 12+ risk rules**; contributed to FastAPI, React, database design, and Docker deployment. Achieved a **Top 20 team position at TetraTHON 2026**.

ClearContract | *Gemini API, React, Chrome Extension, Node.js, Express, MongoDB* July 2026 – Present

- Developed a **Chrome extension prototype** that analyses pasted contract and Terms of Service text using Google's Gemini API to identify potentially concerning clauses and explain them in simpler language.
- Built a **Node.js and Express backend** to handle requests between the browser extension, Gemini API, and MongoDB-based application components.
- Implemented the core **extension-to-backend analysis workflow and authentication components** while continuing to develop the dashboard and broader browser-based analysis features.

Meshmerise | *C++, Arduino IDE, STM32, RLS08, TB6612FNG* October 2025 – December 2025

- Designed and built an **autonomous maze-solving robot** using an STM32 microcontroller and RLS08 line sensors for navigation.
- Implemented control logic in Arduino IDE for **line following and left-hand maze solving** across 90-degree mazes.
- Recorded turns during the first traversal and used them to improve the second traversal through **path optimization**.
- Integrated the **TB6612FNG motor driver** and regulated power delivery using a buck converter.

Technical Skills

Languages: Python, C/C++, SQL, Swift, HTML/CSS, Javascript/Typescript
AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, NumPy, pandas, Matplotlib
Frameworks: FastAPI, React, Node.js, Express
Databases & Tools: PostgreSQL, MongoDB, Git, Github, Docker, VS Code, Arduino IDE, LaTeX